Practice workbook · Sign and magnitude numbers

Signed integers held in 8-bit sign and magnitude: pattern to denary, denary to pattern, and the lowest/highest storable values.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-02

1. Question 2 · part b · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part b only.

2 Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

(2)

(ii) Identify the number of characters that can be represented using standard ASCII.

(1)

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

(2)

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

(2)

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P 7 2 5 3 8 A 0 4 2 0

1. Question 2 · part b · 2 marks

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Worked-example retry · Attempt part b only.

(c) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	1	0	0
0	0	0	1	0	1	1	0

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

(2)

Colour	Binary pattern
Green	
Black	
White	
Red	
Blue	

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1. Question 2 · part b · 2 marks · Working and answer

Working and answer

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2. Question 3 · part e.i · 2 marks

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Further practice · Attempt part e.i only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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2. Question 3 · part e.i · 2 marks

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Further practice · Attempt part e.i only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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2. Question 3 · part e.i · 2 marks

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Further practice · Attempt part e.i only.

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(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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2. Question 3 · part e.i · 2 marks · Working and answer

Working and answer

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3. Question 2 · part a.i · 2 marks

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Further practice · Attempt part a.i only.

2 Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

(2)

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

(2)

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

(1)

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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3. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 2 · part d.ii · 2 marks

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Further practice · Attempt part d.ii only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

4. Question 2 · part d.ii · 2 marks

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Further practice · Attempt part d.ii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

4. Question 2 · part d.ii · 2 marks

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Further practice · Attempt part d.ii only.

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(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

(2)

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

(2)

(Total for Question 2 = 13 marks)

4. Question 2 · part d.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 2 marks

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Further practice · Attempt part b only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

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2 .....

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5. Question 4 · part b · 2 marks

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Further practice · Attempt part b only.

(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

(2)

	8-bit binary pattern	Denary value
Lowest value		
Highest value		

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5. Question 4 · part b · 2 marks

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Further practice · Attempt part b only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

O	S	P	O AND S	NOT P	(O AND S) AND (NOT P)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

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(Total for Question 4 = 13 marks)

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5. Question 4 · part b · 2 marks · Working and answer

Working and answer

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